Stress and strain in pseudo-particle solids

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Abstract

A method is described for computing the stress and strain fields in pseudo-particle solids.

1 Introduction

The discrete element method (DEM) can be used for modeling soil and granular materials. The nonsmooth dynamics approach allows for fast and stable simulation with strong coupling between DEM materials and mechatronics systems modeled using rigid multibody dynamics. In most cases the DEM particles do not represent the true particles, whose size often range to submicron scale. Instead, the DEM particles are considered to be *pseudo-particles* that represent a finite part of the material. The pseudo-particle shape and contact model do not necessarily match the true particle shape and contact mechanism, and are determined empirically from physical experiments and matching numerical experiments for the *bulk mechanical properties.* This is referred to as model calibration [4].

At the macroscopic level (or bulk level) a particle based material appear solid and it is meaningful to estimate continuous fields of mass density, displacement, velocity, strain and stress. These fields may be more easy to visualize and interpret. They can also be compared to what is obtained from continuum soil mechanics by analytical or numerical means. Another motivation is to identify constitutive laws for continuum models of granular media, see [5] for a review of the many alternatives.

The are several mehods for transitioning from a microscopic to a macroscopic description by sampling and averaging the local particle state, also referred to as *homogenization*. The classical Irving-Kirkwood approach [7] of calculating discrete stress fields within small volumes or bins has been generalized with the concept of *Representative Volume Element* (RVE) [1], which include also the strain tensor field. The most popular technique is the so called *coarse-graining* method [6, 14, 13, 3, 10, 11, 12]. The main advantage of the latter is that the fields automatically satisfy the conservation equations of continuum mechanics. It is also review nicely in the textbook [2].

2 Particle level-of-detail

Each particle in the granular system, indexed a, has a position $\mathbf{q}^a = [\mathbf{x}^a, \mathbf{e}^a]$, velocity $\dot{\mathbf{q}}^a = [\mathbf{v}^a, \boldsymbol{\omega}^a]$, mass matrix $\mathbf{M}^a = \mathrm{diag}(m^a, m^a, m^a, \mathbf{I}^a)$ and diameter d^a. The total number of particles is denoted N_p.

The contact force network consist of N_c contact points, indexed n, with contact position $\mathbf{x}_{\mathrm{c},n}$ and pairwise contact force $\mathbf{f}^{ab}_n = -\mathbf{f}^{ba}_n$ that typically consist of a normal force and a tangential friction force, and sometimes also rolling resistance.

For spherical particles the normal force $\mathbf{f}^{ab}_{\mathrm{n},n}$ is directed along the particles center of line and result in no torque. A tangential force $\mathbf{f}^{ab}_{\mathrm{t},n}$ acting at point $\mathbf{r}^a_n$ relative to the center of body a produce a torque $\boldsymbol{\tau}^{ab}_{\mathrm{t},n} = \mathbf{r}^a_n \times \mathbf{f}^{ab}_{\mathrm{t},n}$. The rolling resistance is a pure torque $\boldsymbol{\tau}^{ab}_{\mathrm{r},n}$ with no associated force. The total force and torque on particle a from a contact with particle b is thus $\mathbf{f}^{ab}_n = \mathbf{f}^{ab}_{\mathrm{n},n} + \mathbf{f}^{ab}_{\mathrm{t},n}$ and $\boldsymbol{\tau}^{ab}_n = \boldsymbol{\tau}^{ab}_{\mathrm{t},n} + \boldsymbol{\tau}^{ab}_{\mathrm{r},n}$.

3 Coarse-graining

Coarse graining is a projection process from microscopic resolution to macroscopic description where some subresolution information is lost. The macroscopic scale is defined by a *coarse graining scale*[1], or *smoothing length* R relating to the particle size d. The macroscopic fields are cumpetd usung a *coarse-graining function*, or *smoothing kernel* $\phi(\boldsymbol{x})$. If the smoothing length

[1]Temporal coarse-graining may also be applied, with sone temporal scale.

is chosen too fine ($R \ll d$), the fields will have fluctuations that reflecting the particle nature and force localization in strong force chains. You might or might not be interested in that information. If the smoothing length is too coarse ($R \gg d$)information of macroscopic gradients are lost, e.g., strain localization. A common choice for studying the bulk properties is $R \simeq 1.5d$.

It is assumed that each contacting pair of particles share only on contact (convex particles), each contact can be considered a point contact and that collisions are not instantaneous. Furthermore, most strain tensor definitions assumes small deformations (or even small displacements?).

There are several choices for smoothing kernels $\phi(\boldsymbol{x})$. In general, the choice of smoothing length is more important than the precise kernel function. One requirement is that is hould be normalizable, $\int \phi(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}^3 = 1$, such that the integral of the mass density equals the total mass. A gaussian $\phi(\boldsymbol{x}) = (\sqrt{2\pi}R)^{-3} \exp(-|\boldsymbol{x}|^2/2R^2)$ with cutoff at $|\boldsymbol{x}| = 3R$ is a convenient choice.

The continuous fields are then constructed by sampling the microscopic variables from the particles in the neighbourhood of a coordinate point $\boldsymbol{x}$. The *mass density* field become

$$\rho(\boldsymbol{x}, t) = \sum_a m^a \phi(\boldsymbol{x} - \mathbf{x}^a(t)) \tag{1}$$

The *momentum density* vector field is

$$\boldsymbol{p}(\boldsymbol{x}, t) = \sum_a m^a \mathbf{v}^a \phi(\boldsymbol{x} - \mathbf{x}^a(t)) \tag{2}$$

From the momentum and density fields we can compute the macroscopic *velocity* vector field

$$\boldsymbol{v}(\boldsymbol{x}, t) = \boldsymbol{p}(\boldsymbol{x}, t)/\rho(\boldsymbol{x}, t) \tag{3}$$

The velocity field represent the mean flow. The particle's indivudual motion fluctuate about this velocity due to collisions. In the solid regime the level of fluctuations is small, in the liquid regime it is moderate, and in the gaseous regime the fluctuations are the essence of the state. It is common to quantify the level of fluctuations by the mean square velocity fluctuations, referred to as the *granular temperature*. We can thus construct a granular temperature field

$$T(\boldsymbol{x}, t) = \sum_a \left[\mathbf{v}^a - \boldsymbol{v}(\boldsymbol{x}, t)\right]^2 \phi(\boldsymbol{x} - \mathbf{x}^a(t)) \tag{4}$$

The total stress tensor[2] $\sigma_{\alpha\beta} = \sigma^{\mathrm{k}}_{\alpha\beta} + \sigma^{\mathrm{c}}_{\alpha\beta}$ is the sum of the *kinetic stress tensor*

$$\sigma^{\mathrm{k}}_{\alpha\beta}(\boldsymbol{x}, t) = -\sum_a m^a v^a_\alpha v^a_\beta \phi(\boldsymbol{x} - \boldsymbol{x}^a(t)) \tag{5}$$

and the *contact stress stensor* [3]

$$\sigma^{\mathrm{c}}_{\alpha\beta}(\boldsymbol{x}, t) = -\sum_n f^{ab}_{\alpha,n} x^{ab}_\beta \int_0^1 \phi(\boldsymbol{x} - \mathbf{x}^a(t) + s\mathbf{x}^{ab}(t))\mathrm{d}s \tag{6}$$

where the summation is over the set of contacts, $\mathbf{f}^{ab}_n$ is the contact force between particle a and b, with branch vector $\mathbf{x}^{ab} = \boldsymbol{x}^b - \boldsymbol{x}^a$. The evaluation of the integral depend on the particular choice of kernel function.

From the stress tensor we can compute various quantities such as the *pressure*

$$p = -\tfrac{1}{3}\mathrm{tr}(\boldsymbol{\sigma}) \tag{7}$$

the *stress deviator* tensor

$$\bar{\sigma}_{\alpha\beta} = \sigma_{\alpha\beta} + p\delta_{\alpha\beta} \tag{8}$$

and the *von Mises stress*

$$\sigma_{\mathrm{vm}} = \sqrt{\tfrac{3}{2}\bar{\sigma}_{\alpha\beta}\bar{\sigma}_{\alpha\beta}} = \sqrt{\tfrac{3}{2}\mathrm{tr}(\bar{\boldsymbol{\sigma}}^2)} \tag{9}$$

In order to compute the displacement field and strain tensor field one must track the *particle displacement*

$$\mathbf{u}^a(t) = \mathbf{x}^a(t) - \mathbf{x}^a(0) \tag{10}$$

The *displacement field* is[4]

$$\boldsymbol{u}(\boldsymbol{x}, t) = \frac{\sum_a m^a \mathbf{u}^a \phi(\boldsymbol{x} - \mathbf{x}^a(t))}{\rho(\boldsymbol{x}, t)} \tag{11}$$

[2]The spatial components of the tensor is indexed by greek indices $\alpha, \beta, \ldots$ ranging $1, 2, 3 = x, y, z$. The Einstein summation convention is assumed.

[3]In [6] the coarsegrained stress tensor is derived from the equations of momentum balance, Newton's third law, symmetrization and integration by parts. If the summation is carried out over particles there is a factor 1/2 that is not present when summing over contacts. Luding [8] derives a stress tensor from the relation between deformation energy and stress, $\boldsymbol{\sigma} = \partial U/\partial \boldsymbol{\varepsilon}$.

[4]From a formal derivation going from particle representation to Lagrangian field representation and finally this Eulerian field representation it is clear that this is valid only for linear strains [14].

and the *deformation gradient* become [14]

$$F_{\alpha\beta}(\boldsymbol{x},t) \equiv \frac{\partial u_\alpha}{\partial x_\beta} = \frac{\sum_a \sum_b m^a m^b (u^a_\alpha - u^b_\alpha)(\partial_\beta \phi^a)\phi^b}{\rho(\boldsymbol{x},t)^2} \tag{12}$$

where $\phi^a \equiv \phi(\boldsymbol{x} - \mathbf{x}^a(t))$ and $\partial_\beta \phi^a = \frac{\partial \phi^a}{\partial x_\beta}$. The linear *strain tensor* is finally given by

$$\varepsilon_{\alpha\beta}(\boldsymbol{x},t) \equiv \frac{1}{2}\left(\frac{\partial u_\alpha}{\partial x_\beta} + \frac{\partial u_\alpha}{\partial x_\beta}\right) = \frac{1}{2}\left(F_{\alpha\beta} + F_{\beta\alpha}\right) \tag{13}$$

Similarly the *rate of strain tensor* $\dot{\varepsilon}_{\alpha\beta}(\boldsymbol{x},t)$ become

$$\dot{\varepsilon}_{\alpha\beta}(\boldsymbol{x},t) = \frac{1}{2}\left(\tilde{F}_{\alpha\beta} + \tilde{F}_{\beta\alpha}\right) \tag{14}$$

with the *gradient of the velocity field*

$$\tilde{F}_{\alpha\beta}(\boldsymbol{x},t) \equiv \frac{\partial v_\alpha}{\partial x_\beta} = \frac{\sum_a \sum_b m^a m^b (v^a_\alpha - v^b_\alpha)(\partial_\beta \phi^a)\phi^b}{\rho(\boldsymbol{x},t)^2} \tag{15}$$

In the case of a dense flow with small changes in the mass density, $\nabla\rho \approx 0$, we can approximate the gradients by

$$F_{\alpha\beta}(\boldsymbol{x},t) \equiv \frac{\partial u_\alpha}{\partial x_\beta} \approx \frac{\sum_a m^a u^a_\alpha \partial_\beta \phi^a}{\rho(\boldsymbol{x},t)} \tag{16}$$

$$\tilde{F}_{\alpha\beta}(\boldsymbol{x},t) \equiv \frac{\partial v_\alpha}{\partial x_\beta} \approx \frac{\sum_a m^a v^a_\alpha \partial_\beta \phi^a}{\rho(\boldsymbol{x},t)} \tag{17}$$

The *inertial number* is an important quantity for characterizing the state of a granular systen. It is defined

$$I(\boldsymbol{x},t) = \frac{\dot{\varepsilon}(\boldsymbol{x},t) d}{\sqrt{p(\boldsymbol{x},t)/\rho(\boldsymbol{x},t)}} \tag{18}$$

where the norm of the strain rate is $\dot{\varepsilon} = (\frac{1}{2}\varepsilon_{\alpha\beta}\varepsilon_{\alpha\beta})^{1/2}$ and d is the particle grain size. The inertial number can be interpreted as the ratio of two time-scales for particle rearrangements: the microscopic time-scale that depend on pressure and particle sizes and inertia, and the macroscopic time-scale linked to the shear rate. The quasi-static regime is $I \lesssim 10^{-3}$ [9], the dense liquid regime $10^{-3} \lesssim I \lesssim 10^{-1}$ and the gaseous regime is $I \gtrsim 10^{-1}$.

4 Implementation

An implementation for computing the macroscopic fields from particle data involves chosing a grid and kernel functions, see Fig. 1.

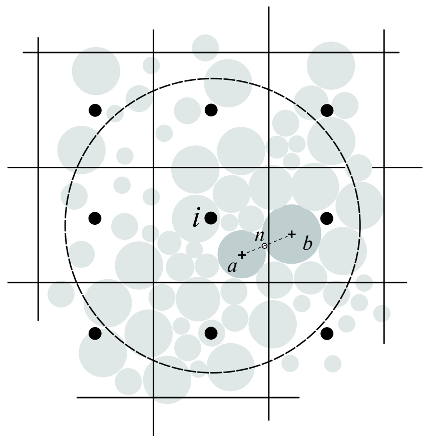

Figure 1: Illustration of coarse graining of particles with diameter d into fields discretized in a voxel grid with sidelength L using smoothing kernels with radius R . Voxels are indexed i and have width L, particles a and b and the contacts by n.

4.1 Grid and particle lists

Introduce a regular grid with cubic cells, voxels, indexed by i and having side length L. The center point of each voxel is denoted $_i$.

All fields $\psi(\boldsymbol{x})$ are descretized and the value at each discrete voxel center $i \in = 1, 2, \ldots, N_\mathrm{v}$ is computed

$$\psi_i = \psi(\boldsymbol{x})_{|\boldsymbol{x}=\boldsymbol{x}_i} \tag{19}$$

All particles within a smoothing length R are involved in the computation. All other particles have zero contribution as the smoothing kernel evaluates to zero. Therefore, to each voxel there need to be list of the particles within it and a list of neighbouring voxels with center points within a distance of R, such that the particles in these voxels also can be accessed.

4.2 Kernel function derivative

If a Gaussian kernel function (with cutoff at $|\boldsymbol{x}| = 3R$) is used we have

$$\phi^a \equiv = \phi(\boldsymbol{x} - \mathbf{x}^a) = \frac{1}{(\sqrt{2\pi}R)^3} e^{-\frac{|\boldsymbol{x}-\mathbf{x}^a|^2}{2R^2}} \tag{20}$$

with derivative

$$\partial_\beta \phi^a = \frac{\partial \phi^a}{\partial x_\beta} = -\tfrac{1}{R^2}(x_\beta - x^a_\beta)\phi^a \tag{21}$$

An alternative is to use Lucy polynomials to obtain fields that are integrable with compact support and differentiable.

4.3 Contacts to stress

The stress field computations involve the contact forces. Therefore, to each voxel there need to be list of the contact points in it or within a distance R from the voxel center. Alternatively this can be reached through the particles but double counting of the contact forces must then be avoided.

The evaluation of the stress is simplified by using the Heaviside kernel function, $\phi(\boldsymbol{x}) = H(R-x)H(R-y)H(R-z)$, in Eq. (6). The stress tensor become [6, 8]

$$\sigma^{\mathrm{c}}_{\alpha\beta}(\boldsymbol{x}_i) = -\frac{1}{V_R} \sum_{n \in \mathcal{N}_R(\boldsymbol{x}_i)} f^{ab}_{\alpha,n} x^{ab}_\beta \tag{22}$$

where the summation is over contacts $\mathcal{N}_R(\boldsymbol{x}_i)$ with center point, $\mathbf{x}_n = \mathbf{x}^a + (d^a/2)\mathbf{x}^{ab}/|\mathbf{x}^{ab}|$, within a box with side-length $2R$ of $\boldsymbol{x}_i$, $\mathbf{x}^{ab} = \mathbf{x}^b - \mathbf{x}^a$, and $V_R = 8R^3$ is the kernel volume. This estimation of stress is valid when the voxel discretization and smoothing length is large compared to the size of the interacting particles [6].

There is no mentioning in the literature on coarse-graining involving rolling resistance. A natural choice is to compute an effective *rolling traction*, $\mathbf{f}^{ab}_{\mathrm{r},n}$, as follows

$$\boldsymbol{\tau}^{ab}_{\mathrm{r},n} \to \mathbf{f}^{ab}_{\mathrm{r},n} = \boldsymbol{\tau}^{ab}_{\mathrm{r},n} \times \frac{\mathbf{x}^{ab}_n}{|\mathbf{x}^{ab}_n|}\frac{2}{d} \tag{23}$$

and include that in the contact force when computing the stress tensor by Eq. (22).

5 Test results

The dependency on R, L and d should be analyzed. A default choice is $R = L = 1.5d$, but the stress is not well resolved with the approximation of Eq. (22). The choice $R = 2L = 1.5d$ might be more suitable for that.

6 Conclusions

Appendix

Particle system equation of motion

$$\mathbf{M}\ddot{\mathbf{q}} = \mathbf{F}_{\mathrm{c}} + \mathbf{F}_{\mathrm{ext}} \tag{24}$$

where $\mathbf{F}_{\mathrm{c}} = [\mathbf{f}_{\mathrm{c}}, \boldsymbol{\tau}_{\mathrm{c}}]$, $\mathbf{f}^{ab}_{\mathrm{n},n} \propto \delta^{3/2}\mathbf{n}$, $|\mathbf{f}^{ab}_{\mathrm{t},n}| \leq \mu_{\mathrm{t}}|\mathbf{f}^{ab}_{\mathrm{n},n}|$ and $|\boldsymbol{\tau}^{ab}_{\mathrm{r},n}| \leq \mu_{\mathrm{r}}\frac{d}{2}|\mathbf{f}^{ab}_{\mathrm{n},n}|$

Solid equation of motion

$$\rho[\partial_t + \boldsymbol{v} \cdot \boldsymbol{\nabla}]\boldsymbol{v} = \boldsymbol{\nabla} \cdot \boldsymbol{\sigma} + \rho \boldsymbol{f}_{\mathrm{ext}} \tag{25}$$

$$\partial_t \rho + \boldsymbol{\nabla} \cdot [\rho \boldsymbol{v}] = 0 \tag{26}$$

with mass density $\rho(\boldsymbol{x})$, velocity $\boldsymbol{v}(\boldsymbol{x})$, stress $\boldsymbol{\sigma}(\boldsymbol{x}) = \boldsymbol{C}\boldsymbol{\epsilon}(\boldsymbol{x})$, $\boldsymbol{\sigma} = \boldsymbol{C}\boldsymbol{\epsilon}$, displacement $\boldsymbol{u}(\boldsymbol{x})$

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